

1 **Three-dimensional data of wire-cut surface scans**
2 **under the confocal microscope**

3 Yuhang Lin^{1,2,*}, Heike Hofmann^{2,3}, Curtis Mosher⁴, Eden Amin⁵, Jeff Salyards²,
4 Alicia Carriquiry^{1,2}

5 ¹Department of Statistics, Iowa State University, USA

6 ²Center for Statistics and Applications in Forensic Evidence, Iowa State University, USA

7 ³Department of Statistics, University of Nebraska-Lincoln, USA

8 ⁴Roy J. Carver High Resolution Microscopy Facility, Iowa State University, USA

9 ⁵Forensic Science Institute, University of Central Oklahoma, USA

10 ^{*}Corresponding author: Yuhang Lin (yhlin@iastate.edu)

11 **Abstract**

12 Wire-cut striation marks provide evidence for forensic source identification, but pub-
13 licly available datasets and reproducible quantitative workflows for analyzing these marks
14 are limited. We present a dataset of 120 three-dimensional topographic scans in **x3p** for-
15 mat from aluminum wires cut with five KWS-105 wire cutters. The study contains six
16 scans—two replicates at each of three cutting locations—for each of the 20 tool-edge com-
17 binations. We provide additional scan metadata, manually extracted profiles, processed
18 signals, cross-correlation function (CCF) values for pairwise aligned signals, images, and
19 analysis code. The accompanying workflow covers scan inspection, profile and signal extrac-
20 tion, alignment, pairwise similarity visualization, CCF distributions for same-source and
21 different-source comparisons, and variability assessment across replicates, stagings, and ac-
22 quisitions. The dataset and workflow support the development, evaluation, and reproducible
23 comparison of quantitative methods for wire-cut evidence.

24 [Agent: Remove the table of contents before preparing the clean Scientific Data
25 submission. The journal’s Data Descriptor outline does not include a table of
26 contents; this one is included only as an internal review aid.]

27 **Contents**

28	Background & Summary	2
29	Methods	3
30	Scan acquisition	3
31	Signal extraction	4
32	Align signals	5
33	Data Records	8
34	Data Overview	9
35	Technical Validation	10
36	Usage Notes	13
37	Data Availability	14
38	Code Availability	14
39	References	14

Background & Summary

In forensic analysis, marks left at crime scenes play an important role, and forensic examiners are particularly interested in identifying their origin, a process known as the Source Identification Problem in Forensic Science¹. The forensic science community focuses on two distinct problems: the **specific source problem**, in which examiners aim to determine whether a mark was left by a particular tool, and the **common source problem**, in which examiners aim to determine whether two marks were left by the same tool. Currently, accepted practice for both problems relies on visual inspection of physical items under a comparison microscope. Examiners then report their conclusions according to the Theory of Identification², developed by the Association of Firearm and Toolmark Examiners (AFTE), using one of three categories: *identification*, in which the marks are believed to have been made by the same source; *elimination*, in which the marks are believed to have been made by different tools; and *inconclusive*, in which the similarities or dissimilarities between marks are insufficient to support either identification or elimination. This process has been criticized by the National Research Council (NRC)³ and the President’s Council of Advisors on Science and Technology (PCAST)⁴ for its lack of objectivity and absence of quantifiable measurements, such as error rates.

In discussing error rates, Biedermann⁵ differentiates between internal and external perspectives. The external perspective allows general statements based on black-box studies that relate examiners’ conclusions to ground truth; however, it does not consider the evidence in a particular case. Introducing an internal perspective requires evaluating similarity between pairs of specimens with known ground truth. Algorithms allow these similarities to be quantified objectively and reproducibly.

Advances in microscopy have made it possible to move from optical two-dimensional imaging to digital three-dimensional topographic scanning, allowing algorithmic assessment of the similarity between pairs of forensic pattern evidence⁶. This transition has already supported practical advances in bullet and cartridge comparisons, including visual inspection of bullet line scans from three-dimensional topographic images⁷, the use of features extracted by fast Fourier transform (FFT) as the basis of an artificial intelligence system⁸, an automated pilot study based on consecutively matching striae⁹ (a feature also used by some laboratories in casework), and the incorporation of three-dimensional imaging into routine casework at the Alabama Department of Forensic Sciences (ADFS)¹⁰.

When cut wires are found at a crime scene, understanding the characteristics of marks left by bladed tools becomes essential. When a blade cuts a wire, it leaves striations on the surface, as shown in Figure 1. These striations provide a basis for similarity assessment and source identification. Similar scans have supported the development of quantitative methods for analyzing firearms and other toolmark evidence. For example, studies of screwdriver striations have examined key factors affecting comparisons, including angles of attack¹¹, rotational axes¹², and cutting directions¹³. Foundational research on forensic toolmarks has also emphasized data collection and analysis, including the collection and distribution of datasets for bullet and toolmark analysis^{14,15}, numerical methods for improving accuracy and consistency^{16,17}, and techniques for quantifying similarity^{18,19}.

Data collected at crime scenes lack the known ground truth needed to estimate error rates; therefore, controlled studies are needed. In this study, we provide a publicly available dataset of wire-cut scans together with a systematic quantitative processing and comparison workflow. The shared materials include raw scans, metadata, manually extracted profiles, processed signals, pairwise cross-correlation function (CCF) values, derived images, and analysis code.

The Methods section describes sample collection, scanning, profile extraction, signal processing, and CCF-based alignment. Data Records describes the shared files and repository organization. Data Overview presents descriptive summaries of pairwise CCF values. Technical Validation assesses signal-processing and alignment behavior and variability across replicates, stagings,



Figure 1. Microscopic wire-cut striations. Close-up of striations left by a blade on the cut end of a wire.

and acquisitions. Usage Notes provides practical guidance for working with the shared files, and Code Availability identifies the custom processing and validation code. The dataset and workflow may support future quantitative research on wire-cut evidence.

Methods

We used five Kaiweets KWS-105 multipurpose tools with 0.5-inch carbon-steel cutting blades to cut 16-gauge anodized pure-aluminum wire.

While aluminum is not commonly used for electric wiring because of its poor conductivity, we chose aluminum for its nontoxicity and suitability for receiving cutting marks and retaining them during subsequent handling. Pure aluminum has a Brinell hardness of approximately 15 HB, which places it between lead (approximately 5 HB) and copper (approximately 35 HB).

Scan acquisition

The blades of each of the five wire cutters were labeled A, B, C, and D (see Figure 2 (a)). B opposes A; the reverse sides are labeled D for A and C for B. Three locations on each blade were marked (inner, middle, and outer), as shown in Figure 2 (a). Each four-inch wire segment was cut in half, creating two cut sides, AB and CD. The cut surface of the wire is not flat; under the microscope, it reveals a roof-like structure with a ridge separating the markings made by each blade (see Figure 1). The two roof sides of each cut surface were then scanned separately, yielding scans A and B from cut side AB and scans C and D from cut side CD. As an example, the four scans resulting from cut 1 by tool 1 are shown in Figure 2 (c).

We scanned the cut surfaces at 20 $\times$ magnification using a Sensofar confocal microscope, shown in Figure 2 (b). The scans were exported in **x3p** format (ISO 25178-72:2017/Amd 1:2020) with a lateral resolution of 0.645×0.645 micrometers per pixel and a vertical resolution of 0.1 micrometer.

The complete design therefore consists of 5 tools $\times$ 3 locations $\times$ 2 replicates $\times$ 4 cut surfaces = 120 scans.

The naming of each piece included the specific tool, its edge, location, and replicate according

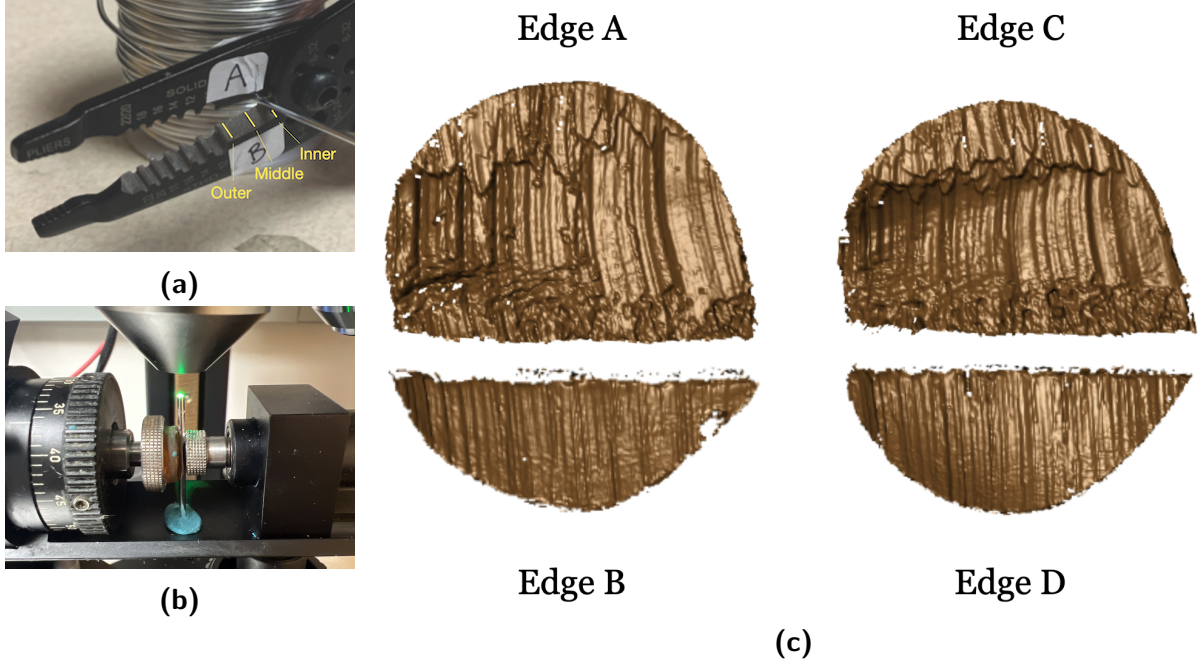


Figure 2. Wire-cut sampling and scanning workflow. (a) A Kaiweets KWS-105 wire cutter with inner, middle, and outer locations marked. (b) A confocal microscope used to scan wire-cut surfaces. (c) Four scans of cutting marks labeled according to the corresponding cutting edges A, B, C, and D.

116 to the following convention:

```

117 T__W-L_-R_
118 || | |
119 || | +-- Replicate: 1 or 2
120 || +----- Location: I (inner), M (middle), or O (outer)
121 |+----- Edge: A, B, C, or D
122 +----- Tool: 1, 2, 3, 4, or 5

```

123 For example, T1AW-LI-R1 denotes the first replicate of a piece cut with edge A of tool 1 at the
124 inner location.

125 To minimize errors during data collection, two team members cut and labeled the wire pieces
126 together. One team member then scanned the wire cuts and assigned names to the scans as
127 described.

128 Signal extraction

129 Signals were extracted from each surface scan using a two-step procedure. First, a representative
130 profile was identified along a line across the surface perpendicular to the dominant striation
131 direction. This line was chosen manually to avoid scanning artifacts (e.g., holes) and cutting
132 artifacts (e.g., rips caused by rolling the wire). Along this line, surface values were assigned at
133 equally spaced positions using the observed surface value at the nearest position on the surface
134 grid.

135 In the second step, we extracted each scan’s signal by removing trends and outliers from the
136 profile’s surface values using two LOWESS filters with bandwidths of 400 μm and 40 μm ,
137 respectively²⁰. All smoothers are susceptible to boundary issues, particularly when outliers are
138 present. Steep changes at the end of a signal have a high potential to attract another signal
139 during alignment, causing misalignments. To avoid potential artificial “hooks” at either end of
140 a signal, we used two-sample t -tests on the surface values in the outermost 2.5% intervals along

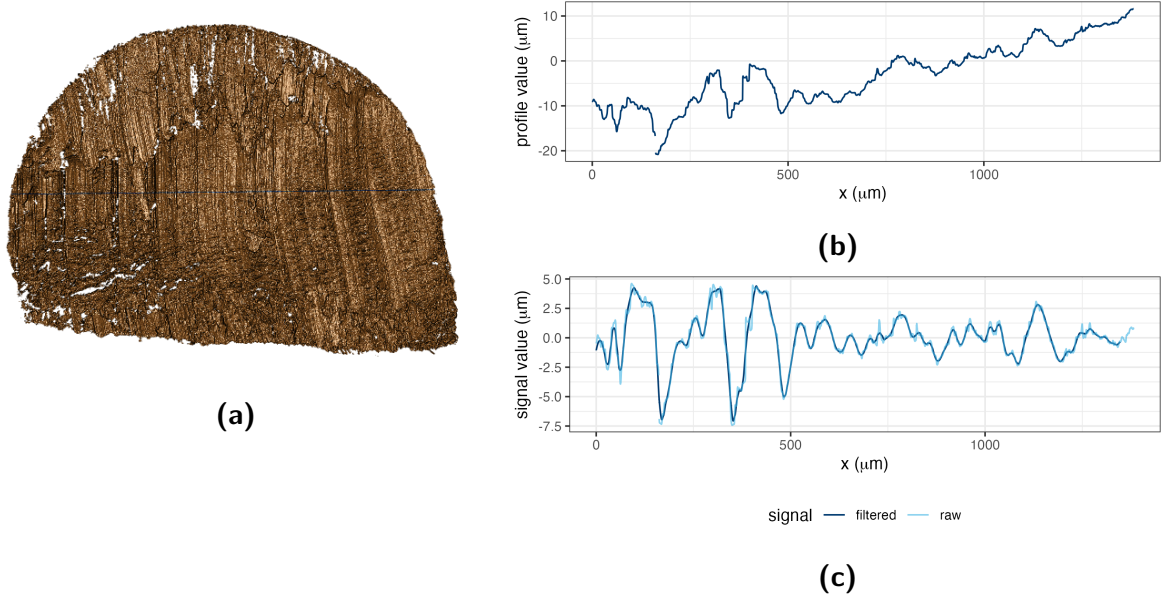


Figure 3. Profile and signal extraction. (a) A dark-blue profile line drawn across the scan striations. (b) The profile function extracted along the line in (a). (c) Raw and filtered signals obtained from the profile function in (b).

the horizontal axis. Specifically, we calculated the summary statistics (sample size, mean, and standard deviation) of the signal values within the intervals $[0, 2.5)\%$, $[2.5, 5)\%$, $(95, 97.5]\%$, and $(97.5, 100]\%$. At each end, the test compared the outermost 2.5% interval with the adjacent interval. We retained the signal values in the outermost interval only when the corresponding p -value was above 0.01, indicating no significant shift.

The widths of these intervals correspond to the choice of bandwidths and the length of the signal. We determined the interval widths using cross-validation.

The resulting signal was used for subsequent comparisons with other signals.

An example profile line is shown in dark blue in Figure 3 (a), and the resulting profile function is plotted in Figure 3 (b).

Figure 3 (c) shows the resulting smoothed signal.

[Heike: do we have a code file? we should refer to that instead of the functions in R we used] [Tom: Yes, but I believe I just call the function on that specific x3p/signal file to get this picture.]

Using `x3ptools::x3p_extract_profile`, we manually drew one line across each scan, perpendicular to the dominant striation direction, and exported the surface-height values and extraction coordinates.

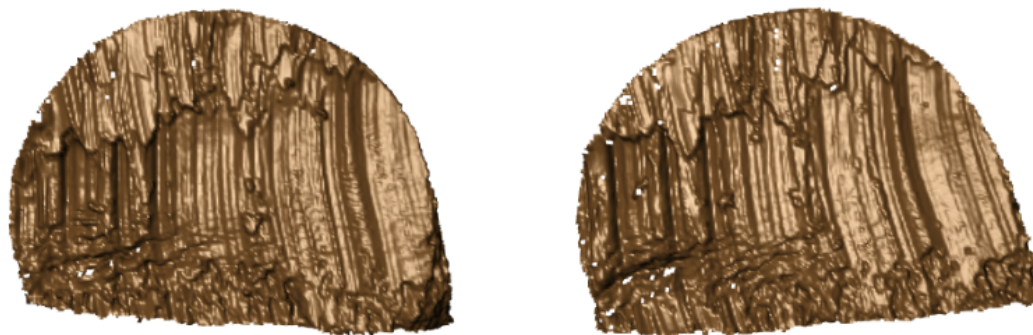
One profile for each scan is included as a CSV file in the repository using the same naming convention as the scans, with the prefix `profiles-`. [Heike: More details on the exact format of these files are provided below.]

[Heike: Code to create signals from profiles is available in `code/4-create_signals_from_profiles.R`]

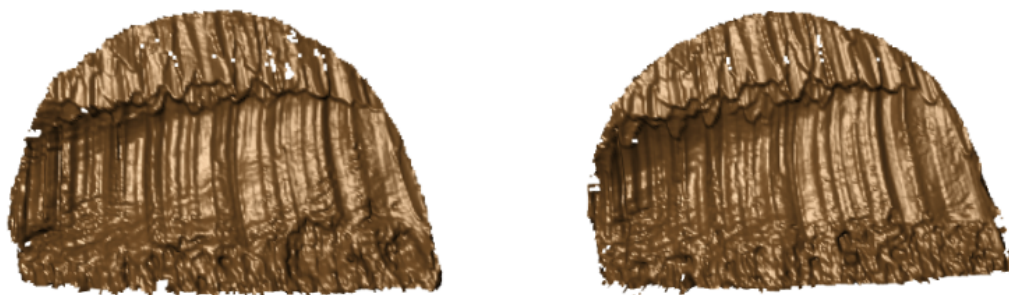
Align signals

After extracting signals from different scans, we compared all possible signal pairs. For each pair, we calculated the CCF across relative shifts and retained the shift with the maximum CCF as the optimal alignment. The comparison between T1AW-LI-R1 and T1AW-LI-R2, for example, corresponds to a row in Figure 4. The first two columns of Figure 5 show the extracted signals, and the rightmost column shows their alignment and CCF value.

Edge A



Edge C



Edge B



Edge D



Figure 4. Same-source replicate scan pairs. Each row shows the two replicate scans for one cut surface (A, C, B, or D) from tool 1 at the inner blade location.

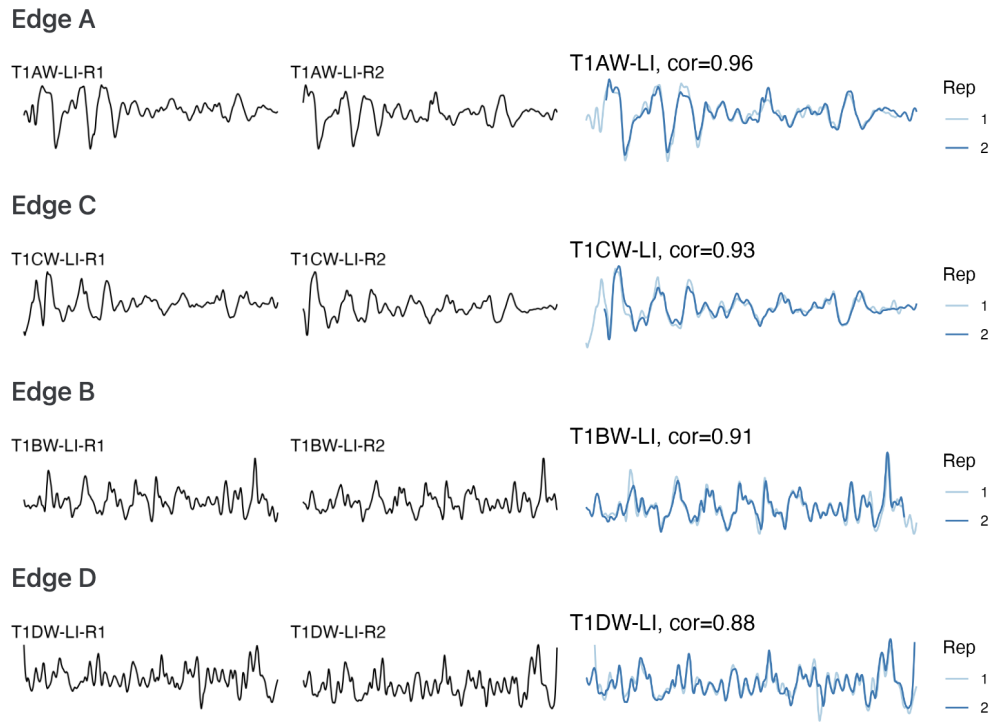


Figure 5. Same-source signal alignment. Rows correspond to cut surfaces A, C, B, and D. The first two columns show signals extracted from the replicate scans in Figure 4; the third column overlays the aligned signals and reports their CCF values.

Data Records

The complete dataset, including the x3p scans, metadata, extracted profiles, processed signals, pairwise CCF values, and derived images, is available from Iowa State University DataShare [DATASET CITATION] at [https://doi.org/\[DOI\]](https://doi.org/[DOI]). [Agent: Replace the dataset citation and DOI placeholders. For first-round review, the destination must permit anonymous download; from the second round onward, it must be a public formal repository record. Add the full dataset reference to References.]

The repository contents, formats, and folder organization are summarized in Table 1.

A data dictionary defining non-obvious columns in meta.csv, the profile and signal files, and the pairwise-CCF file is provided in [repository documentation path]. The dataset is available under the [license] license. [Agent: Verify Table 1 against the deposited record and replace the documentation-path and data-license placeholders. For this non-sensitive original dataset, use CC0 or CC BY rather than a

Table 1. Structure of available data and files.

Data type	Description	Section
Raw data		
scans/	folder containing 120 topographic 3D scans corresponding to 30 aluminum wire cuts (x3p format)	Cutting wires
meta.csv	metadata for each cut, including tool, blade, and location information (CSV format)	Cutting wires
Manual derivatives		
profiles/	folder containing manually extracted profiles (CSV format)	Extract profiles
Computational derivatives in the folder 'data-derived/'		
wire-signals	signals processed from the corresponding profiles (zipped CSV format)	Filtered signals
wire_pairwise_ccf	CCF values for all pairwise aligned signals (zipped CSV format)	Align signals
Image files		
pngs/	folder containing images of 3D wire-cut scans (PNG format)	Cutting wires
profile-images/	folder containing images of profiles extracted from wire cuts (PNG format)	Extract profiles
Visual inventory in the folder 'assessment/analysis-manual/'		
processing-wires	display of pairwise aligned signals from the same sources (HTML format)	Align signals
Technical validation		
variability-assessment/	folder containing technical-validation documents, with an internal structure that mirrors the main folder	Technical Validation

183 license containing NC or SA restrictions.]

184 Data Overview

185 As descriptive summaries of comparison behavior across the dataset, Figure 6 displays pairwise
186 CCF values organized by tool, cut surface, blade location, and replicate. The main diagonal
187 contains self-comparisons, whereas the off-diagonal cells compare distinct signals. Figure 7
188 displays the CCF distributions for same-source and different-source comparisons.

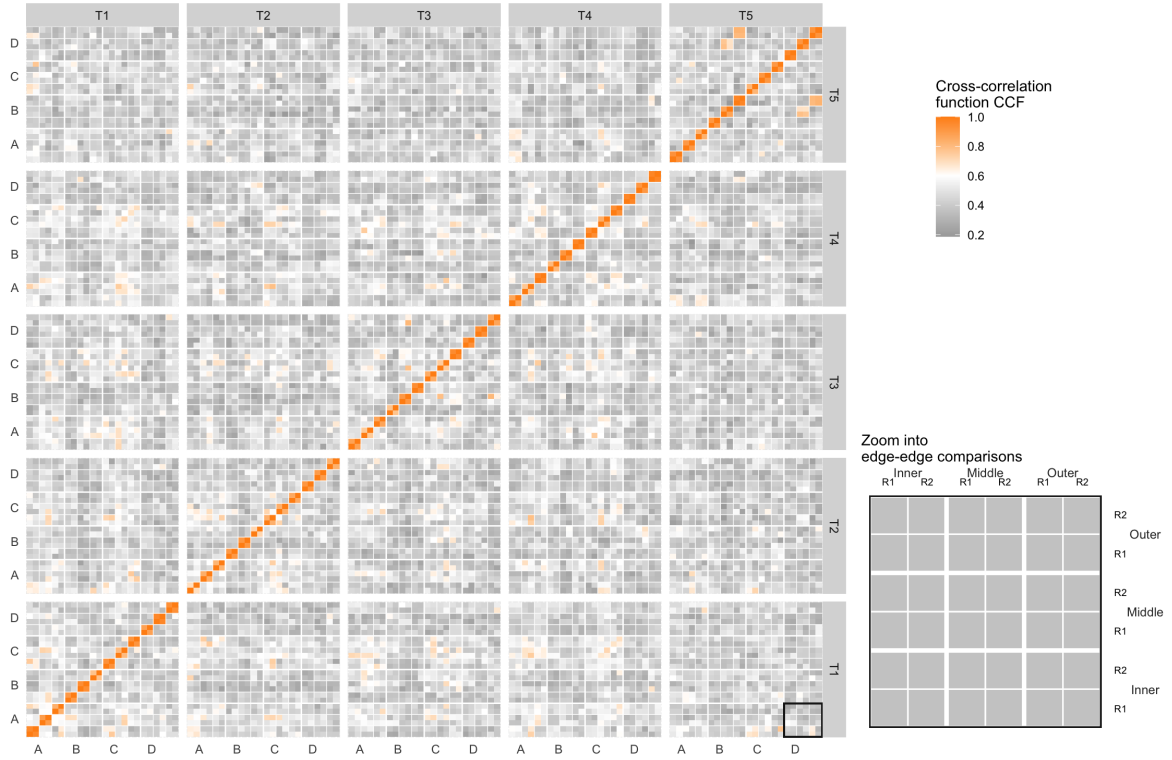


Figure 6. Pairwise CCF tile map. Rows and columns group scans by tool, cut surface, blade location, and replicate. Orange cells indicate larger CCF values and gray cells indicate smaller values; the inset enlarges the boxed edge-to-edge comparison block.

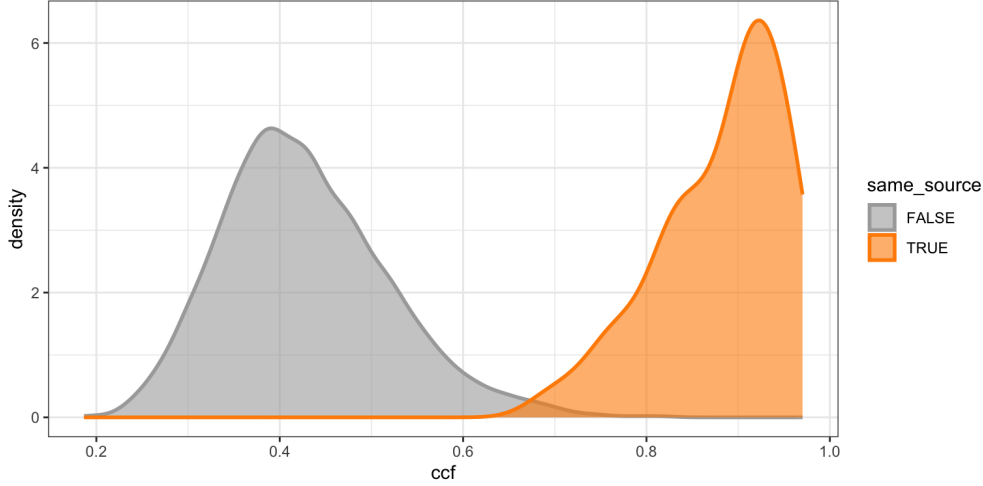


Figure 7. CCF density distributions. Kernel density estimates are shown for different-source comparisons in gray and same-source comparisons in orange.

Technical Validation

We conducted technical validation in two areas: (1) signal processing and alignment and (2) variability assessment.

We validated scan processing and signal alignment as follows. We expected high CCF values between signals from the same source and low values between signals from different sources. Figure 5, shown earlier in Align signals, meets our expectation for same-source signals. As an example of a different-source comparison, we selected T1AW-LI-R1 and T2AW-LI-R1. Figure 8 (a) and (b) show the selected scans. Their signal alignment, shown in Figure 8 (c), has a CCF value of 0.2, which is as low as expected. For the following summaries, edges A and C were categorized as long, edges B and D as short, and comparisons pairing one long edge with one short edge as mixed. We also summarized the resulting CCFs for all pairwise comparisons using boxplots and receiver operating characteristic (ROC) curves. Figure 9 and Figure 10 show these summaries. The ROC curves plot sensitivity, or the true positive rate, against the false positive rate (FPR; $1 - \text{specificity}$). The curves are close to the upper-left corner, indicating strong classification performance. For the combined comparison set, a threshold of 0.589 controls the FPR below 0.05 with a false negative rate (FNR) of 0, whereas a threshold of 0.658 controls the FPR below 0.01 with an FNR of 0.02. These results are consistent with our expectations and validate our pipeline for processing and aligning signals.

The second validation assessed variability in the pipeline across replicates, stagings, and acquisitions. We selected scans from tool 2 at the middle location for this assessment. Replicate variability was assessed using two different cuts at the same edge and position. The comparison in Figure 5 in Align signals illustrates this assessment. To assess staging variability, we rescanned replicate 2 under the confocal microscope and introduced the staging label S into the naming protocol. The S1 designation is omitted from labels, so R2 is equivalent to R2-S1. For all four edges of tool 2 at the middle location, we created three stagings (R2, R2-S2, and R2-S3), computed pairwise CCFs between stagings, and calculated their mean and standard deviation (SD).

Within each staging, we further examined the effect of different acquisitions by keeping the samples on the confocal microscope while varying the lighting conditions. We introduced A as the acquisition label. For edge A at staging 3, we performed four acquisitions and omitted the A1 designation from the first acquisition label. Therefore, R2-S3 is equivalent to R2-S3-A1. For the other three edges, we conducted two acquisitions at staging 3. These acquisitions are denoted by R2-S3 and R2-S3-A2. Within staging 3, we computed pairwise CCFs between

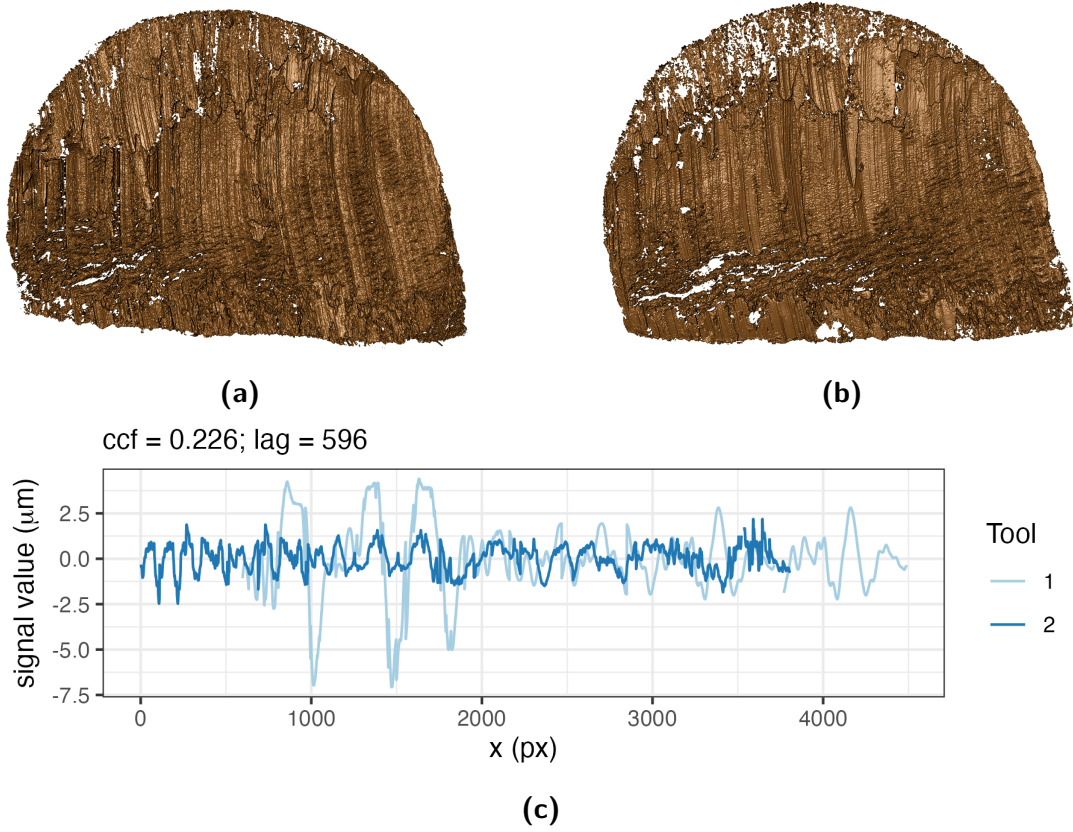


Figure 8. Different-source scan and signal comparison. (a) Scan T1AW-LI-R1 cut by tool 1. (b) Scan T2AW-LI-R1 cut by tool 2. (c) Alignment of signals from T1AW-LI-R1 and T2AW-LI-R1.

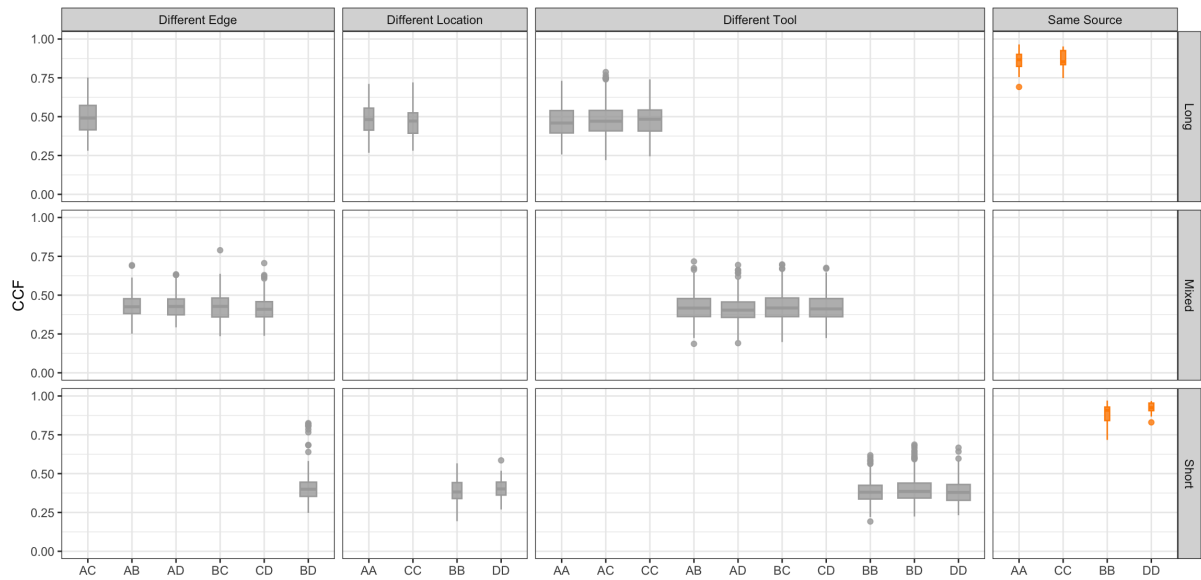


Figure 9. CCF distributions across comparison classes. Gray boxplots show different-source comparisons grouped by different edge, location, or tool; orange boxplots show same-source comparisons. Rows distinguish long, mixed, and short edge comparisons.

222 acquisitions and calculated their mean and standard deviation. Variability between distinct
 223 stagings was assessed separately through the staging comparisons described above.
 224 With this setup, we expected CCF values to become more consistent from replicate to staging

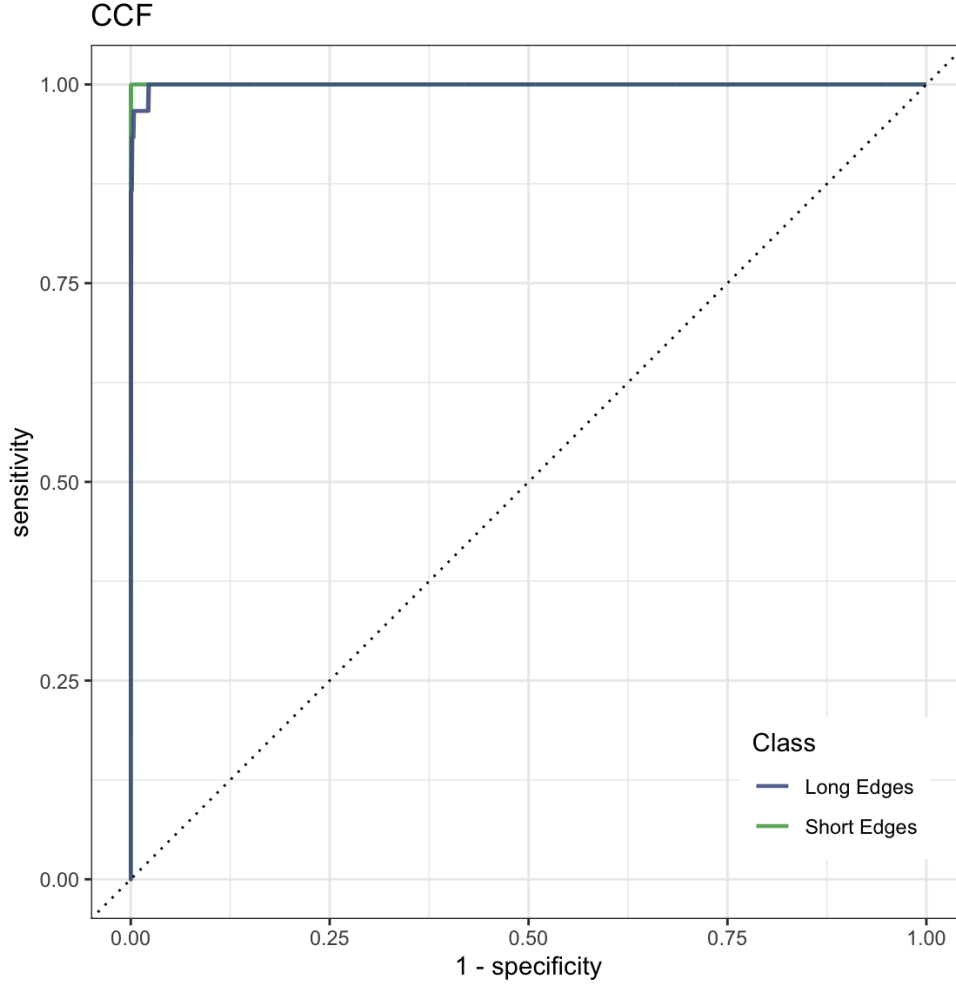


Figure 10. ROC curves for CCF-based classification. Curves are shown separately for long and short edge classes; the dotted diagonal indicates chance-level classification.

to acquisition as the scanning conditions became progressively more stable. Consequently, variability was expected to decrease along this sequence. We also compared replicates, stagings, and acquisitions, expecting smaller differences between acquisitions and stagings than between stagings and replicates. We visualized the results as scatter plots. Figure 11 shows the scatter plots and intervals extending two standard deviations above and below the mean. Higher orange mean lines indicate an increase in the mean CCF, and narrower shaded intervals indicate a decrease in the standard deviation. The detailed numerical results, including means, standard deviations, and differences between means, are shown in Table 2 and confirm our expectations. [Agent: The manuscript contains 11 figures. Scientific Data recommends no more than eight but does not mandate this limit. Confirm with the coauthors or professor whether any logically connected pairs should be combined for revised submission; do not remove a figure or its discussion without an author decision.]

Our variability assessment does not include variation in CCF values caused by differences in the manual extraction of profiles from scans. To support reproduction of the results reported here, we have included the extraction locations and values for each scan.

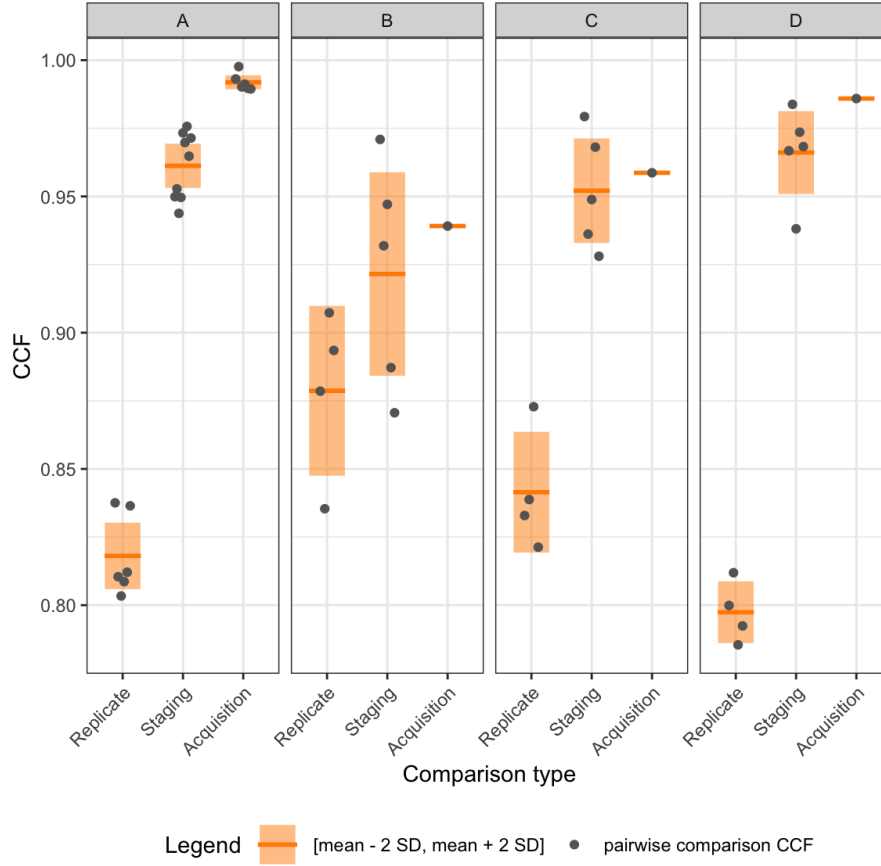


Figure 11. Variability across scanning conditions. Panels A, B, C, and D correspond to the four cut surfaces. Gray points show pairwise CCF values, orange horizontal lines show means, and shaded intervals extend two standard deviations above and below each mean.

Table 2. Comparison of CCF means and standard deviations across different replicates, stagings, and acquisition settings. Dashes indicate that the standard deviation is undefined because only one CCF value is available.

Edge	Replicate (R)		Staging (S)			Acquisition (A)		
	Avg _R	SD _R	Avg _S	SD _S	Avg _S - Avg _R	Avg _A	SD _A	Avg _A - Avg _S
A	0.818	0.015	0.961	0.012	0.143	0.992	0.003	0.031
B	0.879	0.031	0.922	0.042	0.043	0.939	—	0.017
C	0.841	0.022	0.952	0.021	0.111	0.959	—	0.007
D	0.797	0.011	0.966	0.017	0.169	0.986	—	0.020

Usage Notes

The raw surface scans are provided in **x3p** format. To inspect or process these files in R, use the **x3ptools** package and its reference manual, available from CRAN²¹. The **meta.csv** file links each scan to its tool, edge, blade location, and replicate identifiers. The supplied profiles and processed signals provide a starting point for users who do not wish to repeat manual profile extraction. The supplied extraction coordinates allow these derived files to be related back to the corresponding scans. Because each supplied signal is based on a manually selected profile, users assessing sensitivity to profile placement should re-extract profiles from the raw **x3p** scans using alternative documented locations.

Data Availability

The data described in this Data Descriptor are available from Iowa State University DataShare [DATASET CITATION] at [https://doi.org/\[DOI\]](https://doi.org/[DOI]). [Agent: Replace the dataset citation and DOI placeholders with the same record used in Data Records.]

Code Availability

We used custom R code to inspect raw scans, extract profiles, derive signals, align signals, and generate the validation outputs. The code is available in the `assessment/code` folder at <https://github.com/heike/wirecuts-data>. [Agent: For permanent access, archive the exact code release in a DOI-providing repository and cite it if feasible; Scientific Data recommends combining GitHub with a DOI archive.] The files and their roles are described in Table 3.

The provided code reproduces the processing, comparison, validation, and visualization outputs described in Methods and Technical Validation. The workflow was run using R [version] and `x3ptools` [version]. Versions of additional R packages are recorded in [environment file or repository documentation path]. [Agent: Replace the R, `x3ptools`, and environment-record placeholders after checking the analysis environment.]

Table 3. Overview of available code.

Code	Description	Section
Inspect raw scans		
<code>1-create_pngs_from_x3p.R</code>	obtain images of x3p files in <code>scans/</code>	Cutting wires
Extract profiles		
<code>2-create_profiles_from_x3p.R</code>	manually extract profiles from each scan	Extract profiles
<code>3-create-single-profile-file.R</code>	create meta profile information	Extract profiles
Derive signals		
<code>4-create_signals_from_profiles.R</code>	derive signals from each profile	Filtered signals
Align signals		
<code>5-create-images.R</code>	create images for pairwise alignment	Align signals
<code>6-align-pairwise.R</code>	compute pairwise alignment CCF values	Align signals
<code>7-all-comparison-results.R</code>	visualize comparison results	Align signals

References

- [1] Lindley, D. V. A Problem in Forensic Science. *Biometrika* **64**, 207–213, 10.2307/2335686 (1977).
- [2] AFTE. The association of firearm and tool mark examiners: Theory of identification as it relates to toolmarks. *AFTE Journal* **30**, 86–88 (1998).
- [3] National Research Council (U.S.). *Strengthening Forensic Science in the United States: A Path Forward* (National Academies Press, 2009).
- [4] President’s Council of Advisors on Science and Technology. *Forensic Science in Criminal Courts: Ensuring Scientific Validity of Feature-Comparison Methods* (Executive Office of the President of the United States, President’s Council, Washington, D.C., 2016).
- [5] Biedermann, A. The strange persistence of (source) “identification” claims in forensic literature through descriptivism, diagnosticism and machinism. *Forensic Science International: Synergy* **4**, 15 pages, 10.1016/j.fsisyn.2022.100222 (2022).
- [6] Bonfanti, M. S. & Ghauharali, R. I. Visualisation by confocal microscopy of traces on bullets and cartridge cases. *Science & Justice* **40**, 241–256, 10.1016/S1355-0306(00)71994-0 (2000).
- [7] Smith, C. L. Linescan imaging of ballistics projectile markings for identification. In *Proceedings. 36th Annual 2002 International Carnahan Conference on Security Technology*, 216–222, 10.1109/CCST.2002.1049254 (2002).
- [8] Li, D. Ballistics Projectile Image Analysis for Firearm Identification. *IEEE Transactions on Image Processing* **15**, 2857–2865, 10.1109/TIP.2006.877530 (2006).
- [9] Chu, W. *et al.* Pilot Study of Automated Bullet Signature Identification Based on Topography Measurements and Correlations*†. *Journal of Forensic Sciences* **55**, 341–347, 10.1111/j.1556-4029.2009.01276.x (2010).
- [10] McClarin, D. Adding an objective component to routine casework: Use of confocal microscopy for the analysis of 9mm caliber bullets. *AFTE Journal* **47**, 161–170 (2015).
- [11] Baiker, M., Pieterman, R. & Zoon, P. Toolmark variability and quality depending on the fundamental parameters: Angle of attack, toolmark depth and substrate material. *Forensic Science International* **251**, 40–49, 10.1016/j.forsciint.2015.03.003 (2015).
- [12] Garcia, D. L., Pieterman, R. & Baiker, M. Influence of the axial rotation angle on tool mark striations. *Forensic Science International* **279**, 203–218, 10.1016/j.forsciint.2017.08.021 (2017).
- [13] Cuellar, M., Gao, S. & Hofmann, H. An algorithm for forensic toolmark comparisons. *Forensic Science International: Synergy* **9**, 100543, 10.1016/j.fsisyn.2024.100543 (2024).
- [14] Ma, L. *et al.* NIST Bullet Signature Measurement System for RM (Reference Material) 8240 Standard Bullets. *Journal of Forensic Sciences* **49**, 1–11, 10.1520/JFS2003384 (2004).
- [15] Zheng, X. A., Soons, J. A. & Thompson, R. M. NIST Ballistics Toolmark Research Database | NIST (2016).
- [16] Chu, W., Thompson, R. M., Song, J. & Vorburger, T. V. Automatic identification of bullet signatures based on consecutive matching striae (CMS) criteria. *Forensic Science International* **231**, 137–141, 10.1016/j.forsciint.2013.04.025 (2013).

- [17] Vorburger, T. *et al.* Applications of cross-correlation functions. *Wear* **271**, 529–533, 10.1016/j.wear.2010.03.030 (2011).
- [18] Hare, E., Hofmann, H., Carriquiry, A. *et al.* Automatic matching of bullet land impressions. *The Annals of Applied Statistics* **11**, 2332–2356, 10.1214/17-AOAS1080 (2017).
- [19] Ju, W. & Hofmann, H. An Open-Source Implementation of the CMPS Algorithm for Assessing Similarity of Bullets. *The R Journal* **14**, 267–285, 10.32614/RJ-2022-035 (2022).
- [20] Cleveland, W. S., Grosse, E. & Shyu, W. M. Local Regression Models. In *Statistical Models in S* (Routledge, 1992).
- [21] Hofmann, H., Vanderplas, S., Krishnan, G. & Hare, E. "x3ptools: Tools for Working with 3D Surface Measurements", 10.32614/CRAN.package.x3ptools (2018).

Author Contributions

[Agent: Confirm whether all authors approve conversion of this statement to the CRediT taxonomy before revising the contributor roles.]

Y.L.: Methodology, Software, Validation, Data Curation, Writing: Draft; H.H.: Conceptualization, Methodology, Validation, Writing: Review and Editing; C.M.: Laboratory Supervision; E.A.: Physical Specimen, Scanning; J.S.: Forensic Advice; A.C.: Funding Acquisition. All authors reviewed the manuscript.

Competing Interests

H.H. is a technical advisor to the Association of Firearm and Toolmark Examiners (AFTE), a fellow of the American Statistical Association (ASA), and a member of the ASA Forensic Science Committee. H.H. has testified as a court witness on behalf of Judge April Neubauer in the New York State Supreme Court, Criminal Term, in New York City. [Agent: Confirm that the declaration covers every author and replace Alicia's declaration, if any.]

The authors declare the following competing interests:

Funding

This work was partially funded by the Center for Statistics and Applications in Forensic Evidence (CSAFE) through Cooperative Agreement 70NANB20H019 between NIST and Iowa State University. The agreement includes activities carried out at Carnegie Mellon University, Duke University, the University of California, Irvine, the University of Virginia, West Virginia University, the University of Pennsylvania, Swarthmore College, and the University of Nebraska-Lincoln.